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[22AS552]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

SPACE TECHNOLOGY AND ITS APPLICATIONS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. **Missing data may be suitably assumed**

ii. **ANSWER ALL QUESTIONS**

		PART-A	**5 x2=10**		
1.	**a.**	What was the purpose of SITE programme conducted by ISRO in India? Explain the various components of SITE and its significant benefits.	L2	CO1	**02**
	b.	A solar panel of 2m x 2m is irradiated by sun's rays at normal incidence. What is the power generated?	L2	CO2	**02**
	c.	Using the term eccentricity, sketch the orbit shapes for the cases e<1, e = 0 and e >1.	L2	CO3	**02**
	d.	Mention the major constituents of troposphere, stratosphere and ionosphere.	L2	CO4	**02**
	e.	Mention at least two Indian satellites with applications in communication, remote sensing and astronomy.	L2	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Describe the classes of orbits based on: **i.** the altitude, **ii.** the geometry/shape. Give at least one example of missions in each class.	L2	CO1	**05**
		OR			
	b.	State and explain the three laws of Kepler.	L2	CO1	**05**
3.	**a.**	What are the typical altitudes of the following layers in the Earth's atmosphere? **i.** Troposphere **ii.** Stratosphere **iii.** Ionosphere	L2	CO2	**05**
		OR			
	b.	**i.** How are the space debris formed? How are the space missions managed in spite of these debris issues? (2M). **ii.** Describe Kuiper's belt in context with space debris. (2M) **iii.** Sketch the approximate shape of the same and location. (1M)	L2	CO2	**05**
4.	**a.**	List out the various types of rocket propellants. Give one example each of at least two types.(3 + 2 M)	L2	CO3	**05**
		OR			

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	b.	**i.** What are the benefits of multi-staging of rockets? (3 M) **ii.** Using ISRO's PSLV, specify the propellants used in each stage. (2 M)	L2	CO3	**05**
5.	**a.**	List out the various types of attitude sensors and describe any three of them. (2 + 3 M)	L2	CO4	**05**
		OR			
	b.	List out the various types of actuators and describe any three of them. (3 + 2 M)	L2	CO4	**05**
6.	**a.**	Describe how a payload influences on satellite design.	L2	CO5	**05**
		OR			
	b.	What are the major components of an Attitude and Orbit Control Systems (AOCS)? Explain with a complete system diagram.	L2	CO5	**05**
7.	**a.**	**i.** Name the two major classes of thermal control systems/elements. (2M) **ii.** Explain their functional aspects with at least two examples each. (3 M)	L2	CO6	**05**
		OR			
	b.	In order to set right an anomaly in a Mars Orbiting Mission satellite, a telecommand is sent from the Earth ground station. How long will it take to reach the unit on the satellite? A return message is monitored through telemetry. How long will t take to reach the ground station? Consider the average distance between Mars and Earth as 250 million km.	L3	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	From first principles, derive Rocket Equation. Show the various terms in the equation and their significance.	L3	CO1	**10**
		OR			
	b.	**i.** Using a suitable diagram explain the principles and functioning of a rocket engine. (7 M) **ii.** Write a short note on the Rocket nozzle (3 M)	L2	CO1	**10**
9.	**a.**	**i.** What are the various environmental parameters and tests to be considered in a satellite fabrication starting from ON GROUND up to and including operations in space? (8 M) **ii.** Write a short note on Thermo-Vacuum chamber. (2 M)	L2	CO2	**10**
		OR			

	b.	**i.** What are the basic differences between a spacecraft bus and a payload? (3M) **ii.** List out the various subsystems of a spacecraft bus/platform. Explain at least 4 of them.(7M)	L2	CO2	**10**
10.	**a.**	What are the advantages of a space communication as compared to terrestrial communication? Consider merits and demerits of each.	L2	CO3	**10**
		OR			
	b.	What do you understand by the term satellite Transponder? Explain using a block diagram of a BENT-PIPE with all functional parameters	L2	CO3	**10**
11.	**a.**	What are the advantages of remote sensing? Identify the various platforms used in remote sensing missions and describe each of them.	L2	CO4	**10**
		OR			
	b.	List out all the areas of remote sensing applications. Explain at least three of them.	L2	CO4	**10**
12.	**a.**	What are the various astronomy missions conducted globally including those from India? List out at least 5 of them. Explain at least three of the Indian missions, highlighting the scientific goals in each case.	L2	CO6	**10**
		OR			
	b.	**i.** Describe the globally accomplished human-in-space missions as on date. (4 M) **ii.** What are the present plans of India's Gaganyaan Mission? (3 M) **iii.** What were the missions completed by India as a pre-cursor to this? (3 M)	L2	CO6	**10**
13.	**a.**	What are the basic goals and principles of a Global Navigation Satellite Systems? List out the global missions planned / operational at present. Highlight briefly	L2	CO5	**10**
		OR			
	b.	**i.** What is Indian Regional Navigation Satellite System (IRNSS)? (4 M) **ii.** Describe the three segments namely: space segment, ground segment and user segment. (6 M)	L2	CO5	**10**